# Residues and Duality

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# Preface

In the spring of 1963 I suggested to Grothendieck the possibility of my running a seminar at Harvard on his theory of duality for coherent sheaves — a theory which had been hinted at in his talk to the Séminaire Bourbaki in 1957 [**8**], and in his talk to the International Congress of Mathematicians in 1958 [**9**], but had never been developed systematically. He agreed, saying that he would provide an outline of the material, if I would fill in the details and write up lecture notes of the seminar. During the summer of 1963, he wrote a series of "pré-notes" [**10**] which were to be the basis for the seminar.

I quote from the preface of the pré-notes:

> Les présentes notes donnent une esquisse assez détaillée d'une théorie cohomologique de la dualité des Modules cohérents sur les préschémas. Les idées principales de la théorie m'étaient connues dès 1959, mais le manque de fondements adéquats d'Algèbre Homologique m'avait empêché d'aborder une rédaction d'ensemble. Cette lacune de fondements est sur le point d'être comblée par la thèse de Verdier, ce qui rend en principe possible un exposé satisfaisant. Il est d'ailleurs apparu depuis qu'il existe des théories cohomologiques de dualité formellement très analogues à celle développée ici dans toutes sortes d'autres contextes: faisceaux cohérents sur les espaces analytiques, faisceaux abéliens sur les espaces topologiques (Verdier), modules galoisiens (Verdier, Tate), faisceaux de torsion sur les schémas munis de leur topologie étale, corps de classe en tous genres... Cela me semble une raison assez sérieuse pour se familiariser avec le yoga général de la dualité dans un cas type, comme la théorie cohomologique des résidus.
>
> La théorie consiste pour l'essentiel dans des questions de variance: construction d'un foncteur $f^!$ et d'un homomorphisme-trace $\underline{\underline{R}}f_*f^! \to \mathrm{id}$. La construction donnée ici est compliquée et indirecte et n'est pas valable sous des conditions aussi générales qu'on est en droit de s'y attendre. Il faudra sans doute une idée nouvelle pour apporter des simplifications substantielles.

The seminar took place in the fall and winter of 1963–64, with the assistance of David Mumford, John Tate, Stephen Lichtenbaum, John Fogarty, and others, and gave rise to a series of six exposés which were circulated to a limited audience under the title "Séminaire Hartshorne". The present notes are a revised, expanded, and completed version of the previous notes.

I would like to take this opportunity to thank all those people who have helped in the course of this work, and in particular A. Grothendieck, who gave continual support and encouragement throughout the whole project.

R.H.
Cambridge, May 1966

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# Chapter 1

# Introduction

*The way up and the way down are one and the same.*

*–Heraclitus*

The main purpose of these notes is to prove a duality theorem for cohomology of quasi-coherent sheaves, with respect to a proper morphism of locally noetherian preschemes. Various such theorems are already known. Typical is the duality theorem for a non-singular complete curve $X$ over an algebraically closed field $k$, which says that

$$h^0(D) = h^1(K - D),$$

where $D$ is a divisor, $K$ is the canonical divisor, and

$$h^i(D) = \dim_k H^i(X, \mathcal{O}_X(D))$$

for any $i$, and any divisor $D$. (See e.g. [**16**; **Ch. II**] for a proof.)

Various attempts were made to generalize this theorem to varieties of higher dimension, and as Zariski points out in his report [**20**], his generalization of a lemma of Enriques–Severi [**19**] is equivalent to the statement that for a normal projective variety $X$ of dimension $n$ over $k$, for any divisor $D$,

$$h^0(D) = h^n(K - D).$$

This is also equivalent to a theorem of Serre [**FAC**] §76, Thm. 4 on the vanishing of the cohomology group $H^1(X, \mathcal{O}_X(-m))$ for $m$ large and $\mathcal{L}$ locally free. Using a related theorem [**FAC**] §75, Thm. 3, Zariski shows how one can deduce on a non-singular projective variety the formula

$$h^i(D) = h^{n-i}(K - D)$$

for $0 \leq i \leq n$. In terms of sheaves, this result corresponds to the fact that the $k$-vector spaces

$$H^i(X, \mathcal{F}) \quad \text{and} \quad H^{n-i}\big(X, \mathcal{F}^\vee \otimes \omega\big)$$

are dual to each other, where $\mathcal{F}$ is a locally free sheaf, $\mathcal{F}^\vee$is the dual sheaf $\underline{\mathrm{Hom}}(\mathcal{F}, \mathcal{O}_X)$, and $\omega = \Omega^n_{X/k}$ is the sheaf of $n$-differentials on $X$. Serre [**15**] gives a proof of this same theorem by analytic methods for a compact complex analytic manifold $X$.

Grothendieck [**8**] gave some generalizations of these theorems for non-singular projective varieties and then in [**9**] announced the general theorem for schemes proper over a field, with arbitrary singularities, which is the subject of the present lecture notes.

To motivate the statement of our main theorem, let us consider the case of projective space $X = \mathbb{P}^n_k$ over an algebraically closed field $k$. Then there is a canonical isomorphism

$$H^n(X, \omega) \cong k \tag{1.1}$$

where $\omega = \Omega^n_{X/k}$ is the sheaf of $n$-differentials. Combining this with the Yoneda pairing

$$H^i(X, \mathcal{F}) \times \mathrm{Ext}^{n-i}_X(\mathcal{F}, \omega) \to H^n(X, \omega) \tag{1.2}$$

we obtain a pairing

$$H^i(X, \mathcal{F}) \times \mathrm{Ext}^{n-i}_X(\mathcal{F}, \omega) \to k \tag{1.3}$$

which one shows easily to be a perfect pairing [**SGA 62**; **expose 12**]. This generalizes the statements above, because for a locally free sheaf $\mathcal{F}$,

$$\mathrm{Ext}^{n-i}(\mathcal{F}, \omega) = \mathrm{Ext}^{n-i}\big(\mathcal{O}_X, \mathcal{F}^\vee \otimes \omega\big) = H^{n-i}\big(X, \mathcal{F}^\vee \otimes \omega\big).$$

Another way of looking at our duality pairing is as an isomorphism

$$\mathrm{Ext}^{n-i}_X(\mathcal{F}, \omega) \cong \mathrm{Hom}_k\big(H^i(X, \mathcal{F}), k\big). \tag{1.4}$$

Since everything is linear over $k$, we may introduce a $k$-vector space $G$, and have an isomorphism

$$\mathrm{Ext}^{n-i}_X\big(\mathcal{F}, G \otimes_k \omega\big) \to \mathrm{Hom}_k\big(H^i(X, \mathcal{F}), G\big). \tag{1.5}$$

Before proceeding further, we must introduce the derived category. It will be discussed in detail in Chapter I, but for the moment it will be sufficient to know the following: For each abelian category $\mathcal{A}$, there is a category $D(\mathcal{A})$, called the derived category of $\mathcal{A}$, whose objects are complexes of objects of $\mathcal{A}$. If $F\colon \mathcal{A} \to \mathcal{B}$ is an additive functor from one abelian category to another, then under reasonable conditions there is a right derived functor $\underline{\underline{\mathrm{R}}}F\colon D(\mathcal{A}) \to D(\mathcal{B})$ with the property that for any $X \in \mathrm{Ob}\,\mathcal{A}$, if $X$ denotes also the complex which is $X$ in degree zero, and zero elsewhere, then

$$H^i(\underline{\underline{\mathrm{R}}}F(X)) = R^iF(X),$$

where $R^iF$ is the ordinary $i^{\text{th}}$ right derived functor of $F$. Finally, if $F\colon \mathcal{A} \to \mathcal{B}$ and $G\colon \mathcal{B} \to \mathcal{C}$ are two functors then

$$\underline{\underline{\mathrm{R}}}(G \circ F) = \underline{\underline{\mathrm{R}}}G \circ \underline{\underline{\mathrm{R}}}F.$$

This replaces the old-fashioned spectral sequence of a composite functor.

Now we can jazz up our duality for projective space as follows. We replace $k$ by a prescheme $Y$, so that $X = \mathbb{P}^n_Y$. We consider the derived categories $D(X)$ and $D(Y)$ of the categories of $\mathcal{O}_X$-modules and $\mathcal{O}_Y$-modules, respectively. Then cohomology $H^i$ becomes $\underline{\underline{\mathrm{R}}}f_*$, the derived functor of the direct image functor $f_*$, where $f\colon X \to Y$ is the projection. Ext becomes the derived functor $\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}$ of Hom. We define

$$f^!(\mathcal{G}) = f^*(\mathcal{G}) \otimes \omega$$

for $\mathcal{G} \in D(Y)$, and we replace $\mathcal{F}$ by a complex of sheaves $\mathcal{F} \in D(X)$. Then the isomorphism (1.1) gives us an isomorphism

$$\underline{\underline{\mathrm{R}}} f_* \, f^! \mathcal{G} \xrightarrow{\sim} \mathcal{G} \tag{1.6}$$

which we call the trace map.

The Yoneda pairing reappears as a natural map

$$\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_X\big(\mathcal{F}, f^!\mathcal{G}\big) \to \underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_Y\big(\underline{\underline{\mathrm{R}}} f_* \mathcal{F}, \underline{\underline{\mathrm{R}}} f_* \, f^!\mathcal{G}\big), \tag{1.7}$$

which, composed with the trace map (1.6) gives us the duality morphism

$$\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_X\big(\mathcal{F}, f^!\mathcal{G}\big) \to \underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_Y\big(\underline{\underline{\mathrm{R}}} f_* \mathcal{F}, \mathcal{G}\big) \tag{1.8}$$

which generalizes (1.5). This is easily proved to be an isomorphism ([III, 5.1] below) under suitable hypotheses on $Y, \mathcal{F}, \mathcal{G}$. In fact, the proof is nothing but "general nonsense" once one has the isomorphism (1.4).

Having examined the case of projective space, we can state the following ideal theorem, which is the primum mobile of these notes, although it may never appear explicitly in this form.

**Ideal Theorem.**

(a) For every morphism $f\colon X \to Y$ of finite type of preschemes, there is a functor $f^!$ such that

1) if $g\colon Y \to Z$ is a second morphism of finite type, then

$$(gf)^! = f^! \circ g^!;$$

2) if $f$ is a smooth morphism, then

$$f^!(\mathcal{G}) = f^*(\mathcal{G}) \otimes \omega,$$

where $\omega = \Omega^n_{X/Y}$ is the sheaf of highest order differentials;

3) if $f$ is a finite morphism, then

$$f^!(\mathcal{G}) = \underline{\mathrm{Hom}}_{\mathcal{O}_Y}\big(f_*\mathcal{O}_X, \mathcal{G}\big)^{\sim}.$$

# Chapter 2

# The Derived Category

## 2.1 Introduction

## 2.2 Triangulated categories

## 2.3 $\mathcal{K}(\mathcal{A})$ is triangulated

## 2.4 Localization of categories

## 2.5 Quasi-isomorphisms and the derived category

## 2.6 Way-out functors and isomorphisms

## 2.7 Examples. $\mathrm{Ext}$ and $R\mathcal{H}om$

## 2.8 Derived functors

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# Application to Preschemes

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## 3.2 The derived functors of $f_*$ and $\Gamma$

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## 4.3 Recall of the explicit calculations

## 4.4 The trace map for projective space

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## 5.2 Depth and the Cousin complex

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## 7.2 Functorial properties

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# Index of Definitions

# Index of Notations

Of course there is the usual collection of variable notations: $A, B$ for abelian categories or rings, functors or sheaves, $X, Y$ for preschemes, $x, y$ for points of them, $f, g$ for morphisms of preschemes, $\phi$ for morphisms of functors on sheaves, etc.

In general no underline denotes a group, as $\mathrm{Hom}(\mathcal{F}, \mathcal{G})$, one underline denotes a sheaf, as $\underline{\mathrm{Hom}}(\mathcal{F}, \mathcal{G})$ or $\underline{\mathrm{Tor}}(\mathcal{F}, \mathcal{G})$, two underlines denotes an object in a derived category, as $\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}^\bullet(\mathcal{F}^\bullet, \mathcal{G}^\bullet)$, and a dot denotes a complex.

In the list of stable notations below, we have distinguished five categories: Latin alphabet, other alphabets, superscripts, subscripts, and arbitrary symbols.

## Latin alphabet

| Symbol | Meaning | Reference |
|---|---|---|
| Ab | the category of abelian groups | |
| acc | ascending chain condition | V.2 |
| Ass | associated primes | |
| B | boundaries of a complex | |
| Coh | category of coherent sheaves | II.1 |
| Coz | category of Cousin complexes | |
| D | dualizing functor | V.0 |
| $\mathcal{D}(\mathcal{A})$ | derived category | I.4 |
| $\mathcal{D}(X)$ | $= \mathcal{D}(\mathrm{Mod}(X))$ | II.1 |
| dcc | descending chain condition | V.2 |
| dim | dimension | |
| Dual | category of dualizing complexes | VI.1 |
| E | associated Cousin complex | IV.2, IV.3 |
| Ext | Ext groups | I.6 |
| $\underline{\mathrm{Ext}}$ | sheaf Ext | II.3 |
| fid | finite injective dimension | I.7.6, II.7.20 |
| FR1–FR5 | axioms of multiplicative systems | I.3 |
| H | cohomology | I.1, IV.1 |
| Hom | Hom groups | I.6 |
| $\underline{\mathrm{Hom}}$ | sheaf Hom | II.3 |
| $i_x$ | closed immersion | II.7 |

| Symbol | Meaning | Reference |
|---|---|---|
| Icz | category of injective Cousin complexes | |
| id | identity map | |
| im | image | |
| $J(x), J(x, x')$ | standard injectives | II.7 |
| $K(A)$ | | II.7 |
| $k(x)$ | residue field of a point $x$ | |
| $K_\bullet(\underline{f}), K^\bullet(\underline{f}; F)$ | Koszul complexes | III.7 |
| ker | kernel | |
| L1–L3 | axioms of direct systems | I.3 |
| $DerL$ | left derived functor | I.5 |
| lffr | locally free of finite rank | II.5 |
| Lno | category of locally noetherian preschemes | III.8 |
| $\mathrm{Mod}(X)$ | category of $\mathcal{O}_X$-modules | II.1 |
| Ob | objects of a category | |
| pfid | pointwise finite injective dimension | V.8 |
| Ptwdual | category of pointwise dualizing complexes | VI.1 |
| Q | functor into derived category | I.3 |
| Qco | category of quasi-coherent sheaves | II.1 |
| Qis | quasi-isomorphisms | I.4 |
| $\underline{\underline{\mathrm{R}}}$ | right derived functor | I.5 |
| Res | category of residual complexes | VI.1 |
| | residue symbol | |
| | residue of a differential | III.9 |
| Spec | spectrum of a ring | VII.1 |
| Supp | support of a sheaf or module | II.7 |
| T | translation functor | I.1 |
| Tor | | |
| Tot | total complex | II.1 |
| Tr | trace map | III.10, VI.4, VII.1 |
| TR1–TR4 | axioms of a triangulated category | I.1 |
| TRA | axioms of trace | III.10.5, VI.4.2 |
| tr. d | transcendence degree | |
| $\mathrm{Tr}_f$ | trace for a finite morphism | III.6 |
| $\mathrm{Tr}_p$ | trace for a projective morphism | III.11 |
| VAR | axioms for $f^!$ or $f^\Delta$ | III.8.7, VI.3.1 |
| Z | cycles of a complex | |

## Greek and other alphabets

| Symbol | Meaning | Reference |
|---|---|---|
| $\gamma$ | trace for smooth morphism | III.3.4, III.11.2, VII.4.1 |

| | | |
|---|---|---|
| $\Gamma$ | global sections of a sheaf | II.1.5 |
| $\delta$ | fundamental local isomorphism | III.7.3 |
| $\eta$ | unit morphism | |
| $\theta$ | duality morphism | III.5, III.6, III.11 |
| $\iota$ | map of a functor to its derived functor | I.5 |
| $\tau_{\leq n}$ | truncated complex | I.7 |
| $\tau_{\geq n}$ | truncated complex | I.7 |
| $\rho$ | residue isomorphism | III.8 |
| $\omega$ | sheaf of differentials | III.1 |
| $\mathcal{O}_X$ | structure sheaf of a prescheme | |
| $\mathfrak{m}$ | maximal ideal of a ring | |
| $\mathfrak{p}$ | prime ideal of a ring | |
| $\mathbb{A}$ | affine space | |
| $\mathbb{P}$ | projective space | |
| $\mathbb{Q}$ | rational numbers | |
| $\mathbb{Z}$ | integers | |

## Subscripts

| Symbol | Meaning | Reference |
|---|---|---|
| $A'$ | as in $K_{A'}(\mathcal{A}), \mathcal{D}_{A'}(\mathcal{A})$ | I.4 |
| $c$ | as in $\mathcal{D}_c(X)$ | II.1 |
| fid | as in $\mathcal{D}(X)_{\mathrm{fid}}$ | I.7 |
| fTd | as in $\mathcal{D}(X)_{\mathrm{fTd}}$ | II.4 |
| Gor | as in $\mathcal{D}(X)_{\mathrm{Gor}(Z^\bullet)}$ | IV.3 |
| $\mathfrak{p}$ | localization at a prime ideal | |
| qc | as in $\mathcal{D}_{\mathrm{qc}}(X)$ | II.1 |
| $x$ | stalk of a sheaf at $x$ | |
| $Z$ | local cohomology | IV.1 |

## Superscripts

| Symbol | Meaning | Reference |
|---|---|---|
| $b$ | $K^b, \mathcal{D}^b$: bounded | I.2, I.4 |
| $i$ | $i$-th term of a complex / cohomology group | |
| $R^iF$ | $i$-th derived functor | |
| $f^Y$ | base change | VI.2 |
| $Z$ | local cohomology | VI.2 |
| $\bullet$ | denotes a complex $(\mathcal{F}^\bullet, X^\bullet)$ | |
| $\bullet\bullet$ | double complex | I.7 |
| $*$ | opposed category / inverse image $f^*$ | I.7, II.4 |

| | | |
|---|---|---|
| $+$ | bounded below $(K^+, \mathcal{D}^+, R^+)$ | I.2, I.4, I.5 |
| $-$ | bounded above $(K^-, \mathcal{D}^-, L^-)$ | I.2, I.4, I.5 |
| $!$ | $f^!$ (twisted inverse image) | Intr., III.8, VII.3 |
| $\flat$ | $f^\flat$ | IX.6 |
| $\Delta$ | $f^\Delta$ | VI.3 |
| $\vee$ | dual sheaf $(L^\vee)$ | II.5.16, III.1 |
| $-$ | closure $(\lvert x\rvert^-)$ | |
| $\sim$ | sheaf associated to a module | |
| $\wedge$ | completion of local ring | |

## Arbitrary Symbols

| Symbol | Meaning | Reference |
|---|---|---|
| $\lvert$ | restriction | |
| $\gg$ | enough larger than | |
| $\prod$ | product | |
| $\otimes$ | tensor product | |
| $\oplus$ | direct sum | |
| $\coprod$ | disjoint union / sum | |
| $[\,]$ | shift operator / closed interval | I.1 |
| $[\,]$ | bibliography / chapter reference | |
| $\cup$ | union | |
| $\{\,\}$ | the set of | |
| $\to$ | morphism | |
| $\mapsto$ | effect of a map on elements | |
| $+$ | distinguished side of triangle | |
| $\dashrightarrow$ | morphism being constructed | |
| $\circ$ | composition of functors $(F \circ G)$ | I.2 |

# Cohomologie à Support Propre, et Construction du Foncteur $f^!$

by P. Deligne

# Errata